

1. Administrative & Study Identification

Full Study Title: A PHASE 1/2 RANDOMIZED STUDY TO EVALUATE THE SAFETY, TOLERABILITY, IMMUNOGENICITY, AND IMMUNOPERSISTENCE OF A CLOSTRIDIoidES DIFFICILE VACCINE ADMINISTERED WITH NOVEL ADJUVANTS IN HEALTHY ADULTS **Short Title/Acronym:** To Understand the Safety and Effects of a C. Difficile Vaccine With New Adds-Ons That Will Be Given to Healthy Adults **Protocol Number:** C4771001 **NCT Number:** NCT05805826 **Sponsor Name:** Pfizer **Investigational Product:** C. difficile vaccine formulations (Formulation 1, Formulation 2, Formulation 3) and previously studied C. difficile vaccine formulation **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of novel formulations of a *Clostridioides difficile* (C. diff) vaccine administered to healthy adults. **Secondary Objectives:** To determine the optimal dosing schedule (e.g., 0- and 1-month or 0- and 2-month), evaluate immune persistence over time, and assess the safety and immunogenicity of a booster dose.

2.2 Background & Rationale To address the need for a preventive strategy against severe diarrhea and pseudomembranous colitis caused by *C. difficile* infection (CDI). This study explores whether giving new *C. diff* vaccine formulations (incorporating novel adjuvants like CpG or LiNA) helps people mount a more rapid rise in protective antibodies with maintained durability compared to a previously studied *C. diff* vaccine formulation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Saline) and Active Comparator (previously studied formulation) **Blinding:** Observer-Blind / Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Not publicly disclosed (Multiple cohorts across Phase 1 and Phase 2) **Number of Sites:** Multicenter (e.g., NYU Langone Health, Hope Research Institute, DM Clinical Research, etc., United States) **Estimated Study Start:** 23-Mar-2023 **Estimated Primary Completion:** ~2026/2027 (Long-term follow-up spans up to 6 years)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age requirements: Phase 1 (Healthy adults ≥ 65 to <85 years of age), Phase 2 (≥ 65 years of age), Cohort 4 (50 through 64 years of age). 2. Healthy participants as determined by medical history, clinical assessment, and the judgment of the investigator. 3. Capable of giving personally signed informed consent and willing to comply with all scheduled visits and laboratory tests.

4.2 Exclusion Criteria 1. Prior episode of CDI, confirmed by either laboratory test or diagnosis of pseudomembranous colitis at colonoscopy, at surgery, or histopathologically. 2. Previous receipt of an investigational *C. difficile* vaccine or *C. difficile* mAb therapy. 3. Serious chronic disorder, including history of metastatic malignancy, severe COPD, end-stage renal disease, cirrhosis of the liver, or clinically unstable cardiac disease. 4. Known or suspected immunodeficiency or currently receiving treatment with immunosuppressive therapy (e.g., systemic corticosteroids). 5. Any bleeding disorder or anticoagulant therapy that would contraindicate intramuscular injection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered in a 2-dose regimen (evaluating schedules such as 0- and 1-month or 0- and 2-month) vs. placebo. Phase 2 also includes an optional booster dose evaluation. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Phase 1 involves visits spanning roughly 18 months. Phase 2 spans up to 4 years, extended up to 6 years for the booster stage cohort.

5.2 Concomitant & Prohibited Therapy Permitted: General medications for healthy adults not violating the exclusion criteria. **Prohibited:** Anticoagulants, immunosuppressants (including systemic corticosteroids within 28 days before study intervention), receipt of blood products or immunoglobulins within 6 months before enrollment, and concurrent investigational vaccines/drugs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Hematology/Chemistry Labs
Baseline	Day 0	Randomization, First Dose Administration
Interim Visits	Varies	Phase 1: At least 9 visits over 18 months. Phase 2: 6 to 12 visits over 4 years for safety/immunogenicity blood draws.
End of Study / Booster	Up to Year 6	Administration of booster

dose (for selected Phase 2 participants), long-term immune persistence blood collection. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability profile (adverse events, injection site reactions) and immunogenicity (antibody levels against *C. difficile* toxins) of the novel vaccine formulations. **Measurement Method:** Clinical assessment of adverse events and laboratory evaluation of antibody titers from blood samples drawn at prespecified intervals.

7.2 Secondary & Exploratory Endpoints - Optimal dosing schedule (0- and 1-month vs. 0- and 2-month). - Antibody persistence up to 4 years after the last dose of the primary series. - Immunogenicity and safety of a booster dose.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient diary (local and systemic reactions like injection site pain, fatigue) and HCP assessment at follow-up visits. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for any life-threatening events or hospitalizations. **Data Monitoring Committee (DMC):** Internal/External safety review committee per Pfizer's sponsor guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Population (all randomized subjects receiving at least one dose) and Evaluable Immunogenicity Population. **Sample Size Justification:** Designed to be adequately powered to demonstrate non-inferiority or superiority in antibody titer generation against the previously studied vaccine formulation. **Handling of Missing Data:** Standard handling via Multiple Imputation or specific mixed-effect models for repeated measures (MMRM) for immunogenicity titers.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits by Pfizer or delegate CROs to verify source data and safety compliance. **Audit Trail:** Secure eCRF systems with comprehensive audit trails for data entry and query resolution.

11. Study Identifiers & Governance

Primary ID: C4771001 **Registry Number:** NCT05805826 **Sponsor/Lead Organization:** Pfizer **Study Title:** To Understand the Safety and Effects of a C. Difficile Vaccine With New Adds-Ons That Will Be Given to Healthy Adults **Official Regulatory Title:** A PHASE 1/2 RANDOMIZED STUDY TO EVALUATE THE SAFETY, TOLERABILITY, IMMUNOGENICITY, AND IMMUNOPERSISTENCE OF A CLOSTRIDIODES DIFFICILE VACCINE ADMINISTERED WITH NOVEL ADJUVANTS IN HEALTHY ADULTS

12. Clinical Framework (C. Difficile Specific)

Primary Condition: Clostridioides Difficile Associated Disease (CDI) **Diagnosis Threshold:** N/A (Prophylactic healthy adult vaccine trial) **Medical Methodology:** Prophylactic Vaccination **Optimization Target:** Prevention of severe diarrhea and CDI by eliciting a rapid, durable protective antibody response against *C. difficile* toxins.

13. Study Procedures & Methodology

Management Pathway: Vaccination administered intramuscularly to healthy older adults. **Education Protocol (HCP):** Protocol training, injection site reaction assessment training, and blood collection protocols. **Education Protocol (Patient):** Informed consent details regarding vaccine diary keeping and reporting of potential reactogenicity symptoms. **Quality Audit Mechanism:** Investigator site monitoring and sponsor-level data quality reviews.

14. Observation & Follow-Up Schedule

Total Duration: Up to 18 months (Phase 1), 4 years (Phase 2), or 6 years (Booster cohort) **Onsite Visit Cadence:** Phase 1 entails 9 visits; Phase 2 entails 6-12 visits depending on the cohort. **Remote Monitoring Frequency:** Periodic remote safety check-ins as defined by the protocol. **Checklist Review Interval:** Continuous tracking of AEs during the immediate post-vaccination period.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Insert PI Name]	____	[DD-MMM-YYYY]	Clinical Operations Lead	[Insert Sponsor Rep Name]	____	[DD-MMM-YYYY]
Lead Statistician	[Insert Statistician Name]	____	[DD-MMM-YYYY]				